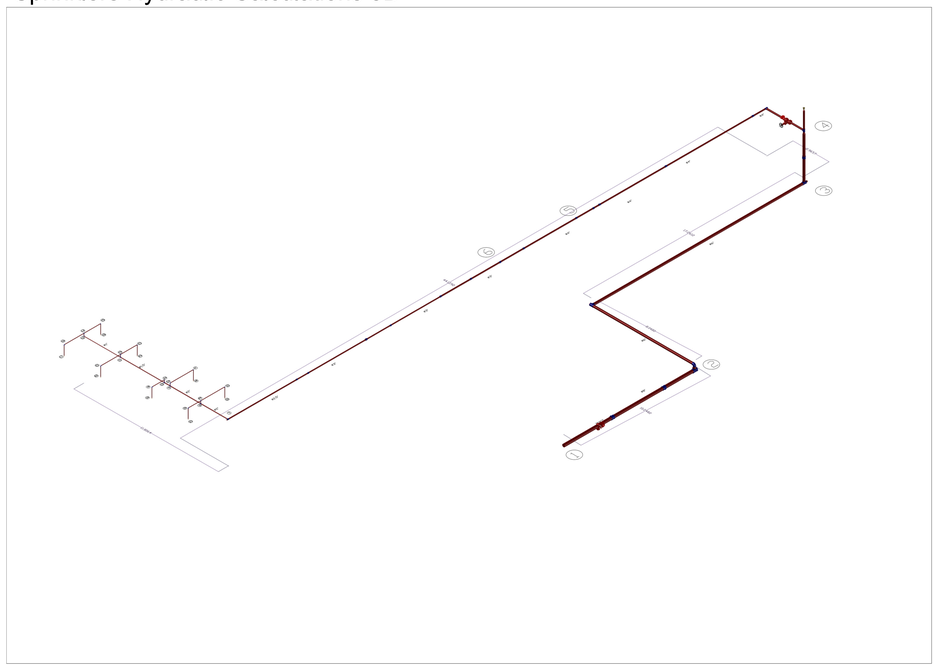
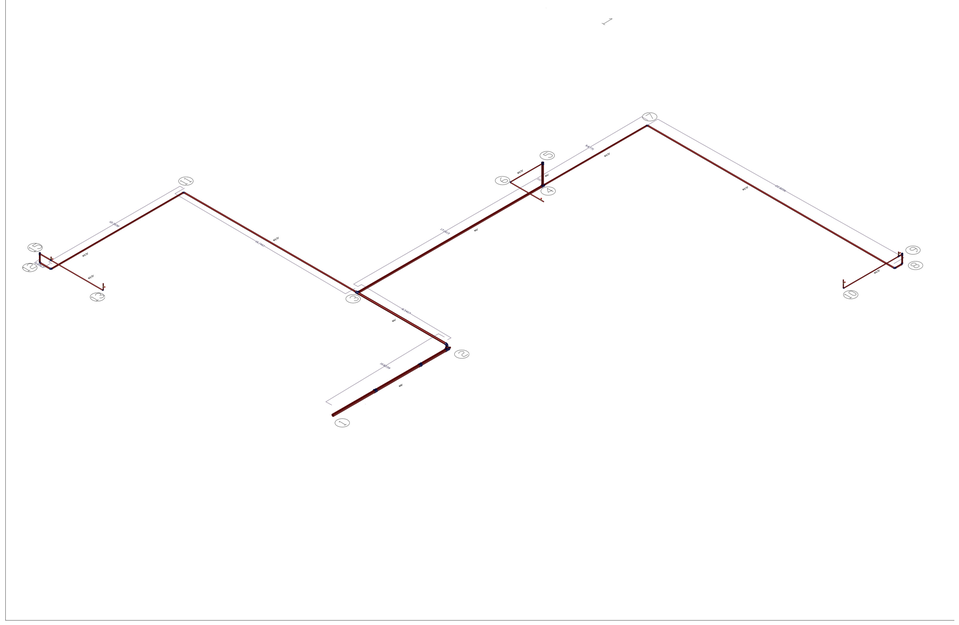
Sprinklers Hydraulic Calculations 3D



Standpipes Hydraulic Calculations 3D



Fire Sprinkler Output Summary

Hydraulically Most Demanding Sprinkler Node

HMD Sprinkler Node Number:	17	
HMD Actual Residual Pressure:	12.76	psi
HMD Actual GPM:	20.00	gpm

Sprinkler Summary

Sprinkler System Type:	Wet	
Specified Area Of Application:	1500.00	ft²
Minimum Desired Density:	0.100	gpm/ft²
Application Average Density:	0.094	gpm/ft²
Application Average Area Per Sprinkler:	250.00	ft²
Sprinkler Flow:	141.64	gpm
Average Sprinkler Flow:	23.61	gpm

Flow Velocity And Imbalance Summary

Maximum Flow Velocity (In Pipe 11 - 19)	17.11	ft/sec
Maximum Velocity Pressure (In Pipe 11 - 19)	1.97	psi
Allowable Maximum Nodal Pressure Imbalance:	0.0100	psi
Actual Maximum Nodal Pressure Imbalance:	0.0073	psi
Actual Average Nodal Pressure Imbalance:	0.0030	psi
Actual Maximum Nodal Flow Imbalance:	-0.4118	gpm
Actual Average Nodal Flow Imbalance:	0.0270	gpm

Overall Network Summary

Number Of Unique Pipe Sections:	32	
Number Of Flowing Sprinklers:	6	
Pipe System Water Volume:	322.63	gal
Sprinkler Flow:	141.64	gpm
Non-Sprinkler Flow:	100.00	gpm
Total System Demand Flow:	241.64	gpm
Minimum Required Residual Pressure At System Inflow Node:	44.04	psi
Demand Flow At System Inflow Node:	241.62	gpm

Fire Hose Output Summary

Hydraulically Most Demanding Hose Node

HMD Hose Node Number:	10	
HMD Actual Residual Pressure:	100.00	psi
HMD Actual GPM:	250.00	gpm

Hose Summary

Hose System Type:	Wet	
Specified Area Of Application:	1500.00	ft²
Minimum Desired Density:	0.100	gpm/ft²
Application Average Density:	0.526	gpm/ft²
Application Average Area Per Hose:	500.00	ft²
Hose Flow:	788.90	gpm
Average Hose Flow:	262.97	gpm

Flow Velocity And Imbalance Summary

Maximum Flow Velocity (In Pipe 5 - 6)	19.03	ft/sec
Maximum Velocity Pressure (In Pipe 5 - 6)	2.44	psi
Allowable Maximum Nodal Pressure Imbalance:	0.0100	psi
Actual Maximum Nodal Pressure Imbalance:	0.0068	psi
Actual Average Nodal Pressure Imbalance:	0.0040	psi
Actual Maximum Nodal Flow Imbalance:	0.0024	gpm
Actual Average Nodal Flow Imbalance:	0.0002	gpm

Overall Network Summary

Number Of Unique Pipe Sections:	12	
Number Of Flowing Hoses:	3	
Pipe System Water Volume:	293.97	gal
Hose Flow:	788.90	gpm
Fixed Flow:	0.00	gpm
Total System Demand Flow:	788.90	gpm
Minimum Required Residual Pressure At System Inflow Node:	145.13	psi
Demand Flow At System Inflow Node:	788.89	gpm

	Sprinklers	Stand Pipes
No.Elements	6 Sprinklers	3 Cabinets
Flow Rate Per One	25 GPM + 100 $GPM_{Allowance}$	250 GPM
Total Flow Rate	250 GPM	750 GPM
Head Static	0.5 bar	0.2 $bar_{just\ one\ hose\ verticaly}$
Head Residual	1 bar	6.9 bar
Head Friction	1.6 bar	2.9 bar
Total Head	3.2 bar	10 bar

- Pumps Selection :
  - Diesel Pump

    Required GPM = 150% of Maximum GPM
    GPM_{Diesel} = 750 x 1.5 = 1125 GPM Nearly to 1200 GPM
    Total $Head_{Diesel}$ = 10 bar

  - ELEC. Pump

    Required GPM = 150% of Maximum GPM
    $GPM_{Electric}$ = 750 x 1.5 = 1125 GPM Nearly to 1200 GPM
    Total $Head_{Electric}$ = 10 bar

  - Jokey Pump

    Required GPM = 10% of Maximum GPM or GPM for one sprinkler
    GPM_{Jokcy} = 750 x 0.1 =75 GPM
    Total $Head_{Diesel}$ = 10 bar + 0.5 bar = 10.5 bar

  - Water Tank

    Volume = GPM x 90 MIN
    Volume = 750 x 60 = 170 m^3 fore each tank = 85 m^3
    Hight Of water = 3.8 m

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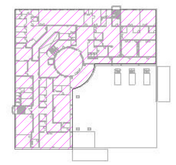
KEY PLAN:



LEGEND:

	Upright Sprinkler		BASE HANGER
	PENDENT SPRINKLES		CHECK VALVE
	SIDEWALL SPRINKLES		PRESSURE GAUGE
	DIESEL PUMP		SUBMERSIBLE PUMP
	ELECTRIC PUMP		DIESEL TANK
	JOCKEY PUMP		FLOW METER
	CONCRETE FOUNDATION		FLOW SWITCH
	CONTROL PANEL		ALARM CHECK VALVE
	BATTERIES		BALL VALVE
	OS&Y VALVE		MUFFLER

Rev.	Date	Detail	Made	Chk'd	App'd

constractor:
Engineer's Company for Electromechanical Works

Consultant :
ASEC For ElectroMechanical

Project :
Menofia Hospital

Title :
Hydraulic Calculations

Eng :	M.Hadhoud	CAD :	ENGX
Chk'd :	Hassan Salah	App'd :	ENGX
Date :	Issue Date	Scale :	As indicated

Drg. No. 08

Rev:

cm

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